

★ WEIGHTED AVG + ALLIGATION ★

Quant | Ch04 Work + Pipes + Speed | Topic 1 of 12

► Meaning in one line

$\text{sum}(w \times x) / \text{sum}(w)$, cheaper/dearer/mean , difference-ratio.

► Key points

Point	Example
$\text{sum}(w \times x) / \text{sum}(w)$.	$60 \times 2 + 90 \times 3 / 5 = 390 / 5 = 78$.
$(\text{dear} - \text{mean}) : (\text{mean} - \text{cheap})$.	12 + 20 for 15: $(20 - 15) : (15 - 12) = 5 : 3$.

★ IMPORTANT FOR EXAM

Replacement + dilution

⚠ TRAP - read twice

Simple-mean + 'alligation always' + ratio-flip

✓ **Try yourself:** Rs.30×2kg + Rs.50×3kg avg? Mix 10 + 25 for 20? **Answer:** $210/5 = 42$; $(25-20):(20-10) = 5:10 = 1:2$.

★ REPLACEMENT + DILUTION ★

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► Meaning in one line

Final = initial $\times ((\text{total} - \text{out}) / \text{total})^n$, water, wine, milk.

► Key points

Point	Example
$F = I \times ((T - o) / T)^n$.	100L, out 10 twice: $100 \times (0.9)^2 = 81\text{L}$ milk.
add-water + ratio + price.	1:3 milk:water, price 1/4th.

★ IMPORTANT FOR EXAM

Ages + work

⚠ TRAP - read twice

Single-step + 'linear' + power-miss

✓ **Try yourself:** 80L, out 20 twice. Left? **Answer:** $80 \times (0.75)^2 = 45\text{L}$.

★ AGE TRANSLATION + RATIOS ★

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► Meaning in one line

Present , past/future , difference-constant , sum , ratio.

► Key points

Point	Example
now x, +n/-n + shift.	A now 20, B 12; 5y ago ratio? 15:7.
difference constant + sum.	ratio 3:5, sum 40 → 15, 25.

★ IMPORTANT FOR EXAM

Family-average + work-rate

⚠ TRAP - read twice

Present-only + 'ratio stays' + difference-shift

✓ **Try yourself:** Father/son 5:1, diff 24. Ages? **Answer:** $5x - x = 24 \rightarrow x6: 30, 6$.

★ FAMILY AVG + WORK-RATE ★

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► Meaning in one line

Total/n , new-born/marriage , shift.

► Key points

Point	Example
total±shift.	avg 5 of 30 → total 150.
combined = $ab/(a+b)$ for 2.	6 + 12 days → $72/18 = 4$ days.

★ IMPORTANT FOR EXAM

Combined + efficiency + men-days

⚠ TRAP - read twice

Average-only + 'add rates' + shift-miss

✓ **Try yourself:** A 10d + B 15d together? Family avg 4 of 20? **Answer:** $150/25 = 6d$; total 80.

★ COMBINED + EFFICIENCY WORK ★

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► Meaning in one line

Together , ratio , parts , wages.

► Key points

Point	Example
$1/a + 1/b + 1/c$; wages ratio.	A+B 12d, wages 9000 in 1:2 → 3000, 6000.
$A = 2B \rightarrow B = 2A$ days.	A twice B, together 12d: A 18d, B 36d.

★ IMPORTANT FOR EXAM

Men-days + alternate + wages

⚠ TRAP - read twice

Add-days + 'equal wages' + efficiency-flip

✓ **Try yourself:** A+B 10d, A–B 1:2 wages 6000? **Answer:** 2000, 4000.

★ MEN-DAYS-HOURS + ALTERNATE DAYS ★

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► Meaning in one line

$M1D1H1/W1 = M2D2H2/W2$, pairs.

► Key points

Point	Example
pairs equal.	$12 \text{ men } 8d = 16 \text{ men } x \rightarrow x = 6d.$
2-day work + remainder.	$A+B \text{ } 2d = 1/6+1/12?$ A6d B12d alternate A-first: $2d = 1/4$; 4 cycles = 8d + ...

★ IMPORTANT FOR EXAM

Wages + pipes

⚠ TRAP - read twice

Men-only + 'alternate = half' + remainder-miss + who-first

✓ **Try yourself:** 20 men 12d = 30 men? A5d B10d alternate B-first, total? **Answer:** 8d; Per 2d (B+A) = $3/10$; 3 cycles = $9/10$ in 6d; day-7 B finishes $\rightarrow$ 7 days.

★ WAGES + INLET RATES ★

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► Meaning in one line

Wages-work-ratio , inverse-days.

► Key points

Point	Example
pay = work ratio.	A 8d B 12d, 5000 → 3000, 2000.
net rate + time.	fill 6h + empty 10h: net $\frac{1}{6} - \frac{1}{10} = \frac{1}{15} \rightarrow 15\text{h}$.

★ IMPORTANT FOR EXAM

Leak + pipes-times

⚠ TRAP - read twice

Equal-pay + 'inlet only' + net-miss

✓ **Try yourself:** A6d B12d 9000? Fill 4h empty 12h net? **Answer:** 6000, 3000; $\frac{1}{4} - \frac{1}{12} = \frac{1}{6} \rightarrow 6\text{h}$.

★ LEAK + PIPES SCHEDULE ★

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► Meaning in one line

Leak = negative-rate , extra-time.

► Key points

Point	Example
leak slows + extra.	fill 8h, leak +2h → net $1/8 - 1/x$... leak rate = $1/8 - 1/10 = 1/40$ → empties 40h.
schedule + left.	A+B 2h, A leaves, B alone 6h → ...

★ IMPORTANT FOR EXAM

Speed + conversion + relative

⚠ TRAP - read twice

Leak-ignored + 'fill rate' + schedule-miss

✓ **Try yourself:** Fill 6h, with leak 8h. Leak alone? **Answer:** $1/6 - 1/8 = 1/24$ → 24h.

★ SPEED + UNIT CONVERSION ★

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► Meaning in one line

$D = S \times T$, $\text{km/h} \times 5/18 = \text{m/s}$, back.

► Key points

Point	Example
$D = ST$.	72 km/h 25 min = $72 \times 25/60 = 30$ km.
$\times 5/18$, $\times 18/5$.	90 km/h = 25 m/s; 20 m/s = 72 km/h.

★ IMPORTANT FOR EXAM

Relative + meeting + average-speed

⚠ TRAP - read twice

Time-units-mix + 'km/h = m/s' + factor-flip

✓ **Try yourself:** 54 km/h 40 min dist? 15 m/s in km/h? **Answer:** 36 km; 54 km/h.

★ RELATIVE + MEETING SPEED ★

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► Meaning in one line

Same/opposite , sum/difference , meet/catch , time.

► Key points

Point	Example
same –, opposite +.	60 + 40 opposite = 100 km/h approach.
meet = dist/sum; catch = dist/diff.	300 km, 50+70 → 2.5 h.

★ IMPORTANT FOR EXAM

Average-speed + trains

⚠ TRAP - read twice

Always-add + 'meet halfway' + direction-miss

✓ **Try yourself:** A60 B90 same-dir behind 120 km. Catch time? **Answer:** Diff 30 → 4 h.

★ AVG SPEED (UNEQUAL) + TRAIN-POLE ★

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► Meaning in one line

Total/total-time , equal-harmonic , train, object/speed.

► Key points

Point	Example
total/total-time.	$60@2h + 90@1h = 210/3 = 70.$
length/speed.	240m train, 72 km/h = 20 m/s $\rightarrow$ 12 s.

★ IMPORTANT FOR EXAM

Platform + two-trains

⚠ TRAP - read twice

Simple-mean + 'train point' + length-miss

✓ **Try yourself:** 30@1h + 90@2h avg? 180m, 54 km/h pole? **Answer:** $210/3 = 70$; 15 m/s $\rightarrow$ 12 s.

★ PLATFORM + TWO TRAINS ★

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► Meaning in one line

Train, platform/speed , opposite/same , sum/difference , meet.

► Key points

Point	Example
$(T+P)/S$.	150+250m, 72 km/h = 20 m/s → 20 s.
$(L1+L2)/(v1\pm v2)$.	120+180m, 60+90 opposite km/h = 41.67 m/s → 7.2 s.

★ IMPORTANT FOR EXAM

Boats + clocks + calendar

⚠ TRAP - read twice

Train-only + 'same = add' + units-mix

✓ **Try yourself:** 200+300m, 90 km/h opposite 54? Cross time? **Answer:** Sum 144 km/h = 40 m/s; 500/40 = 12.5 s.